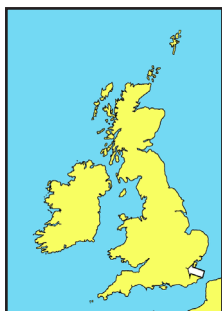


SYNOPSIS



At 2321 on 1 June 2011, the privately owned rigid-hulled inflatable boat *Morfil* collided with the passenger ferry *Sun Clipper* by Blackfriars Road Bridge on the River Thames, London. The vessels were travelling in opposite directions. On impact, *Morfil*'s two crew were pitched into the water but were quickly rescued by the local inshore lifeboat; both were shocked but uninjured. *Morfil* eventually grounded under the road bridge and was a constructive total loss. Damage to *Sun Clipper* was only superficial and there was no pollution.

The investigation identified several factors contributing to this accident, including:

- *Morfil*'s coxswain was under the influence of alcohol and did not take action to avoid *Sun Clipper* until between 1 and 2 seconds before the collision.
- The action taken by *Sun Clipper*'s master to avoid the collision was limited by the proximity of the road bridge and mooring buoys.
- Refurbishment works under the Blackfriars Road Bridge resulted in both vessels using the same bridge arch and their skippers not being able to see each other until about 10 seconds before the collision.
- *Morfil*'s speed was significantly greater than the 12 knot limit recommended by the Port of London Authority.
- *Morfil*'s coxswain had limited knowledge and experience of navigating on the River Thames and was unaware of, or ignored, the local regulations and advice.

There have been at least 45 fatalities resulting from accidents to pleasure vessels over the last 6 years in which alcohol has been a contributory factor. It was extremely fortunate that a further two fatalities did not result from this collision.

The introduction of an alcohol limit for persons in charge of pleasure vessels was first recommended in *The Hayes Report* almost 20 years ago. Although the provision for such a limit was made in the Railways and Transport Safety Act, 2003, the pertinent subsections of the Act have yet to be commenced. The use of byelaws by harbour authorities to deter alcohol consumption on pleasure vessels is largely ineffective.

A recommendation has been made to the Department for Transport aimed at expediting the enactment of a national alcohol limit to persons in charge of pleasure vessels. A recommendation has also been made to the Port of London Authority designed to further enhance the safety of all water users on the River Thames.

SECTION 1 - FACTUAL INFORMATION

1.1 PARTICULARS OF *MORFIL* AND *SUN CLIPPER* AND ACCIDENT

SHIP PARTICULARS

Vessel's Name	<i>Morfil</i>	<i>Sun Clipper</i>
Flag	Not applicable	UK
Classification society	Not applicable	Not applicable
IMO number	Not applicable	9232292
Type	Zodiac Medline II Rigid-hulled Inflatable Boat	Category A, passenger craft high speed catamaran
Registered owner	Private ownership	Thames Clippers
Manager(s)	Not applicable	Thames Clippers
Construction	GRP	Aluminium
Length overall	6.0m	30.03m
Registered length	Not applicable	Not applicable
Gross Tonnage	Not applicable	98gt
Minimum safe manning	Not applicable	3
Authorised cargo	Not applicable	138 passengers

VOYAGE PARTICULARS

Port of departure	St Katherine's Pier, River Thames	Blackfriars Pier, River Thames
Port of arrival	Cadogan Pier, River Thames (intended)	Bankside Pier, River Thames
Type of voyage	Pleasure trip	Commuter service
Cargo information	Not applicable	35 persons
Manning	2	3

MARINE CASUALTY INFORMATION

Date and time	1 June 2011, 2321
Type of marine casualty or incident	Less Serious Marine Casualty
Location of incident	By Blackfriars Road Bridge, River Thames, London

Vessel's Name	<i>Morfil</i>	<i>Sun Clipper</i>
Injuries/fatalities	2 persons overboard	None
Damage/environmental impact	Constructive Total Loss	Superficial damage to fendering
Ship operation	On passage	On passage
Voyage segment	Transit	Transit
Persons on board	2	38
External & internal environment	It was dark. The visibility was good and the river was calm. The flood tide was running at a rate of approximately 3 knots. The water temperature was 11°C. Sunset was at 2107; evening civil twilight was at 2153	

1.2 NARRATIVE

1.2.1 Events preceding the accident

On the evening of 1 June 2011, the co-owners of the rigid inflatable boat (RIB) *Morfil* met unexpectedly during a performance at a London theatre, and agreed to take their boat out for a short trip on the River Thames after the show.

The two men met later at the RIB's permanent berth at Cadogan Pier (**Figure 1**). The co-owner, who normally acted as the RIB's coxswain (herein referred to as 'the coxswain'), remembered that he had removed the lifejackets from the RIB during the previous week, but the two men decided to continue their impromptu trip without them.

At approximately 2230, the coxswain manoeuvred *Morfil* away from her berth and commenced passage downriver towards the Houses of Parliament. He navigated *Morfil* by eye and remained as close as possible to the south riverbank. Both men enjoyed the scenery and the sensation of being out on the river, and decided to extend their trip to St Katherine's Pier (**Figure 1**).

Morfil arrived at St Katherine's Pier at approximately 2244 and, as soon as the RIB was securely moored, the owners went ashore and visited a public house in St Katherine's Dock. They returned to the pier at about 2312.

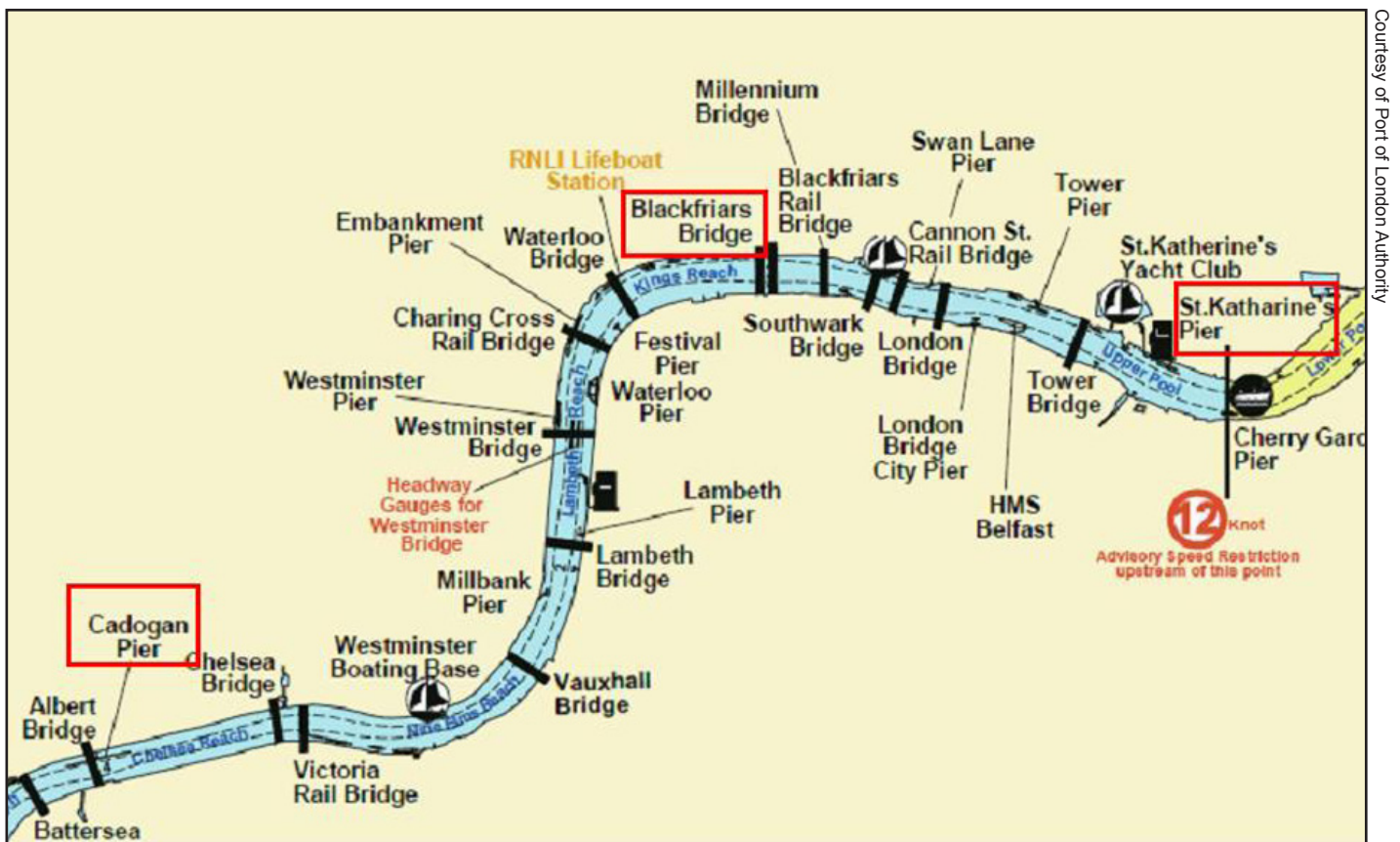


Figure 1: The River Thames

1.2.2 The collision

Shortly after 2312 the coxswain started the engine and switched on the navigation lights. He then cleared the lines and, at about 2316, he manoeuvred the RIB clear of the pier and headed upriver. The coxswain stood behind the steering wheel with the co-owner standing to his left (**Figure 2**). It was a dark but clear night, and the river was calm. The tidal stream was flooding at a rate of approximately 3 knots.

The coxswain increased the engine throttle to about three quarters of full power so that the RIB was travelling on the plane¹ at a speed that he found comfortable to handle. The coxswain kept the RIB close to the bank on his starboard side but did not monitor the boat's progress upriver. *Morfil* passed under several bridges, but the RIB's owners were oblivious to the bridges' names and locations.

At 2319, the high speed craft (HSC) passenger ferry, *Sun Clipper*, departed Blackfriars Pier (**Figure 3**) as scheduled. On board were her three crew and 35 passengers. She was heading downstream towards Tower Bridge on her final service of the day. Her next scheduled stop was at Bankside Pier (**Figure 3**).

Sun Clipper's master checked the Thames Automatic Identification System (AIS) display² to see if any larger vessels were to the east of the Blackfriars bridges and travelling upstream. No vessels were detected in a position to cause concern, so the master manoeuvred *Sun Clipper* towards No.3 arch of the Blackfriars Road Bridge (**Figure 4**). The mate then joined the master on the bridge.

¹ On the plane may be defined as the vessel moving at speed so that its hull is skimming over the water's surface.

² From 1 June 2007 all passenger and commercial vessels on the River Thames, with a length overall greater than 40 metres or over 50gt, were required to carry Thames AIS, which is operated by the Port of London Authority.

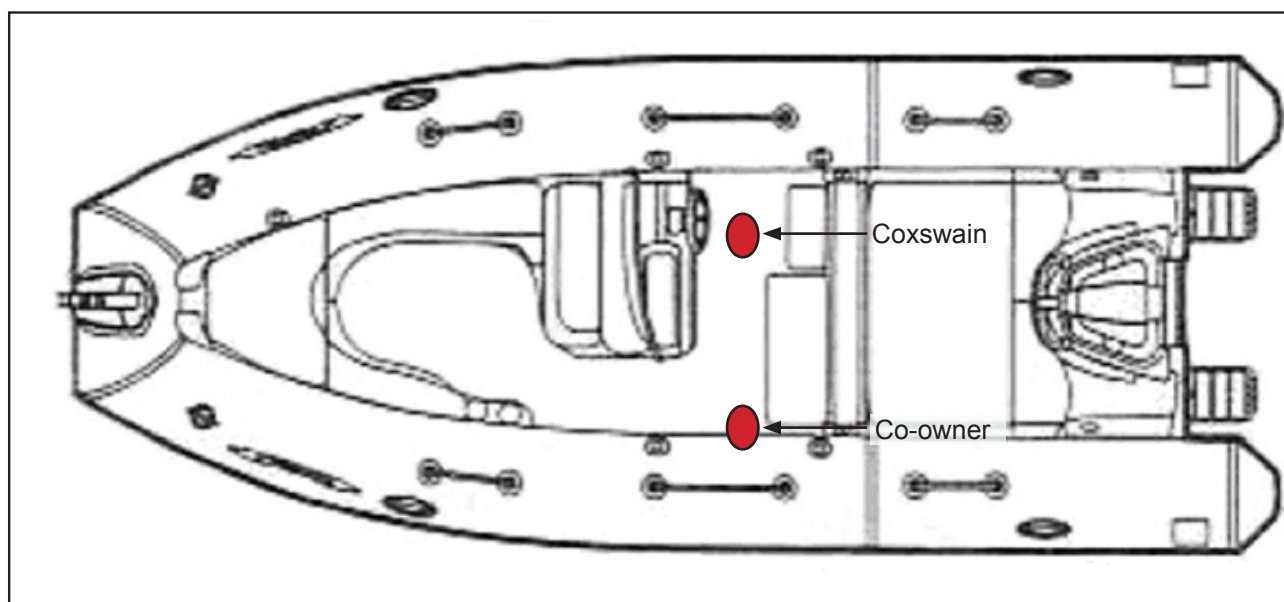


Figure 2: Plan of *Morfil* showing position of occupants

Sun Clipper was approaching No.3 arch at a speed over the ground of 8 knots when the master and mate noticed a small vessel off the port bow, later identified as *Morfil*, heading towards them (**Figure 5**). The RIB was also heading for No.3 arch but from the opposite side. The master and mate saw that *Morfil* was on the plane and estimated that she was travelling faster than 30 knots. *Sun Clipper*'s master immediately realised that a collision was imminent, and reduced the engine throttles to zero.

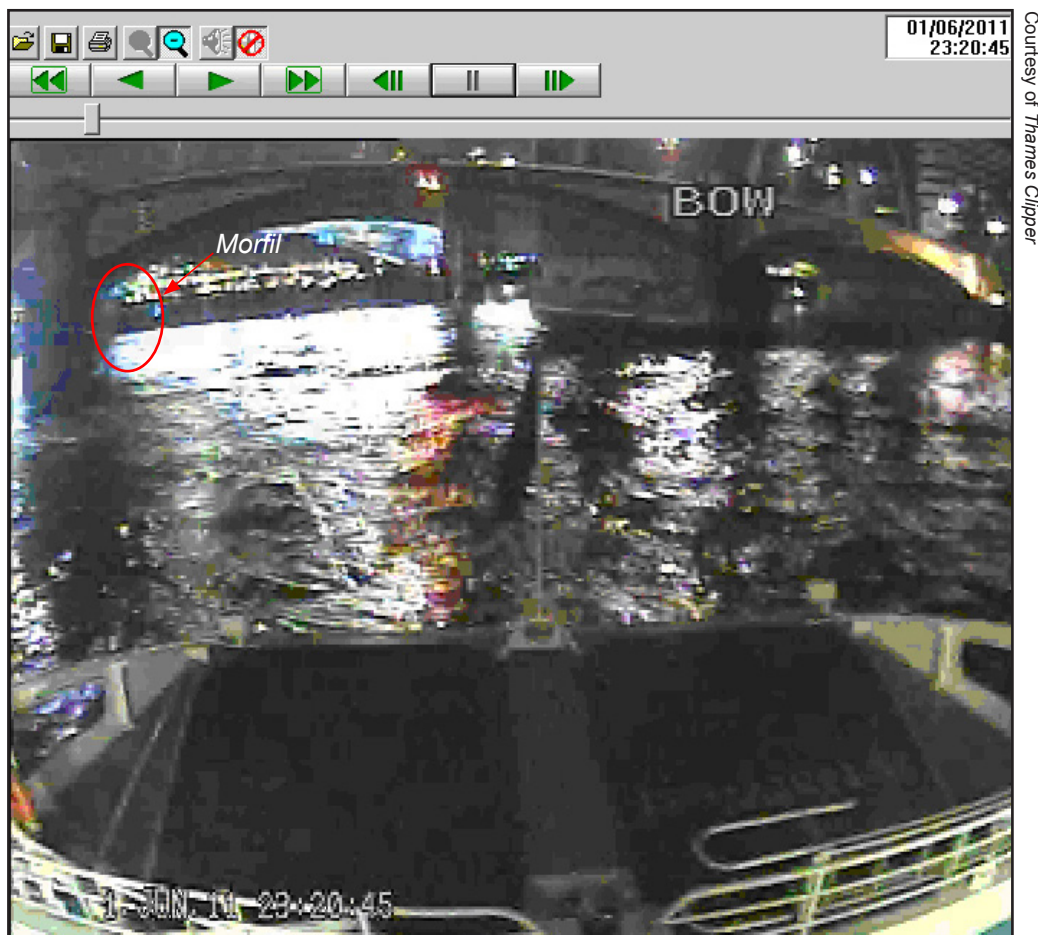


Figure 5: CCTV still - first sighting of *Morfil*

Moments later, *Morfil*'s coxswain saw *Sun Clipper* directly ahead, and turned the steering wheel to port and then to starboard (**Figures 6a to 6c**). This action caused the RIB to swerve violently. Seconds later, at 2321, *Morfil* collided with *Sun Clipper*'s port bow and both the RIB's occupants were thrown overboard.

1.2.3 The rescue

Sun Clipper's master immediately sent the mate forward to look for damage and to see what had happened to the RIB's crew. From the foredeck, the mate saw two people in the water near the HSC's stern. He shouted to the master not to use the engines, and then proceeded to the aft deck. The master acknowledged the mate's warning and then liaised with Thames Vessel Traffic Services (Thames VTS) via Very High Frequency (VHF) radio, channel 14, in order to initiate a rescue.



Figures 6a to c: CCTV still - *Morfil*'s swerve to starboard

Meanwhile, some of *Sun Clipper*'s passengers gathered on the vessel's open aft deck and threw two lifebuoys to the men in the water. The passengers then used the lifebuoy's lines to pull the men to the vessel's side (**Figure 7**).

Morfil, which was now unmanned, continued to turn to starboard at a fast speed until it hit and ricocheted off the stone buttress between the No.1 and No.2 arches of Blackfriars Road Bridge. The RIB then circled back into mid-stream and collided

with *Sun Clipper* for a second time, this time amidships on her port side. *Morfil* then passed close to her two owners in the water before turning towards the road bridge, where the RIB grounded and stopped under No.1 arch.

A Metropolitan Police Marine Policing Unit (MPU) launch and a Royal National Lifeboat Institution (RNLI) inshore lifeboat arrived at the scene at 2325. The RNLI crew had some difficulty manhandling *Morfil*'s owners into the lifeboat; they appeared to be in shock and smelt strongly of alcohol. Neither of the owners was aware of the location of the accident.



Courtesy of Royal National Lifeboat Institution

Figure 7: CCTV still - *Morfil*'s owners at stern of *Sun Clipper*

1.2.4 Subsequent actions

Morfil's owners were recovered from the water and taken ashore to the lifeboat station (**Figure 7**), where they were treated for shock. *Morfil*'s co-owner was later transferred to hospital for observation. The police noticed that the coxswain smelt of alcohol and that his speech was slurred. The coxswain was requested to provide a breath specimen, and he was voluntarily breathalysed at 0220 on 2 June 2011. The breathalyser displayed a "fail" result, which indicated that his breath alcohol level exceeded 35 microgrammes of alcohol in 100 millilitres; the test did not provide the actual level of alcohol in the coxswain's breath.

1.2.5 Damage

Morfil suffered extensive damage (**Figure 8**) and was a constructive total loss. *Sun Clipper* suffered only superficial damage to her fendering.

The risk assessment had not been updated to reflect the use of Blackfriars Bridges' No.3 arches while the No.2 arches were closed. Thames Clippers' vessels did not usually sound one prolonged blast when leaving Blackfriars Pier, which is 125m upstream of Blackfriars Road Bridge, as required by River Byelaw 36.

1.5 THE PORT OF LONDON

1.5.1 River Thames

The River Thames is the longest river in England. It is tidal from its estuary to Teddington Lock and is crossed by 29 bridges along this stretch. Blackfriars Road Bridge spans the River Thames from Blackfriars on the north bank to Southwark on the south bank (**Figure 3**). The road bridge was built in 1869 and has five steel frame arches that are supported by stone faced buttresses (**Figure 9**). Blackfriars Rail Bridge is 48m downriver from the road bridge and also has five arches. The bridges' No.3 arches are normally used by larger vessels; smaller vessels generally use No.2 and No.4 arches.

Courtesy of Port of London Authority



Figure 9: Blackfriars Road Bridge

Refurbishment of Blackfriars Rail Bridge started in March 2009, which required each of the rail and road bridge arches to be closed at various times. Mariners were advised of the current works taking place via the Port of London Authority (PLA) Notice to Mariners (NTM) M3/2011 (**Annex A**), which included:

Persons in charge of vessels are to navigate with particular care and proceed at slow speed when passing the works

On the evening of 1 June 2011, the No.2 arches on both Blackfriars Road and Rail bridges were closed to navigation. The work barge, *Haven Seafield*, and tug *Horton* were positioned under the No.2 arch of Blackfriars Road Bridge. The closure of the No.2 arches on the Blackfriars Road and Rail bridges was indicated by a triangle (apex down) of three red lights in accordance with PLA River Byelaw 29.

1.5.2 Port of London Authority

The PLA was established in 1909 by the Port of London Act and is the statutory harbour authority for the 95 mile tidal stretch of the River Thames from Teddington Lock to the Thames Estuary⁸. The PLA's area of responsibility is divided into two

⁸ For the purpose of this report, future references to the River Thames means the stretch of the river over which the PLA has jurisdiction, namely from the Thames Estuary to Teddington Lock

districts, Upper (from Teddington Lock to Crossness) and Lower (from Crossness to the Thames Estuary). Each of the districts is administered by a harbourmaster, both of whom report to a chief harbourmaster.

The Port of London is the UK's second largest port. In 2010, 39.8 million tonnes of goods were imported through the port and 8.3 million tonnes of goods were exported. Approximately 40 passenger boat operators trade on the River Thames, resulting in around 200,000 passenger pleasure vessel movements carrying 6.5 million passengers each year; about 750,000 people a year commute to work using the regular shuttle services. Over 70 pleasure vessel clubs and marinas are located on the River Thames, in addition to a large number of rowing clubs. The PLA approves and monitors about 300 organised events on the river each year.

1.5.3 Key events in 2012

In 2012, London is hosting two major events that will have a significant impact on the River Thames and its users. The first event is the Queen's Diamond Jubilee in June, which includes a procession of 1000 vessels. The second event, the Olympic Games, is expected to substantially increase vessel activity on the Thames between 27 July and 12 August.

1.6 LOCAL REGULATION

1.6.1 General

The River Byelaws 1978, as amended, specify the requirements for both commercial and pleasure users on the River Thames. The PLA issues NTMs to provide information and advice on works and events as well as changes to rules and regulations which may affect river users. NTMs may be either temporary or permanent. The PLA is also empowered by the Port of London Act to issue navigational instructions to masters of vessels through General Directions⁹.

1.6.2 Alcohol and drugs

River Byelaw 9 states:

The master of a vessel shall not navigate the vessel when unfit by reason of drink or drugs to do so.

The master of a vessel shall not navigate, attempt to navigate or be in charge of a vessel after consuming so much alcohol that the proportion of it in his breath when tested in accordance with paragraph (5) below records a reading of 35 microgrammes of alcohol or more in 100 millilitres of breath.

If the harbourmaster has reasonable cause to suspect that the master of a vessel has drugs or alcohol in his body which may impair his fitness to navigate, he may direct the vessel to proceed to a designated berth or mooring or, if already on a berth or mooring, to remain in that position.

⁹ A vessel's master may choose not to comply with a General Direction if he can prove that he has reasonable grounds that compliance would imperil his vessel or that compliance was impractical.

The harbourmaster may permit a vessel to proceed notwithstanding that the master is suspected of being unfit to navigate through drink or drugs, if the harbourmaster considers that satisfactory arrangements have been made to replace the said master and to ensure safe navigation.

A vessel directed under paragraph (3) above shall remain in the position designated until such time as either a substitute master is on board and takes command of the vessel or the master suspected of having alcohol in his body submits to a breath test on equipment provided by the harbourmaster and approved by the Secretary of State for the purpose of the Road Traffic Act 1988 and the said breath test indicates a reading of less than 35 microgrammes of alcohol in 100 millilitres of breath.

It is an offence for the master of a vessel to fail to comply with a direction made under paragraph (3) above.

The PLA has initiated 5 successful prosecutions in the last 12 years, following alcohol-related accidents. Of these, two were, inter alia, against River Byelaw 9. Notwithstanding these successful prosecutions, the PLA has received legal advice that River Byelaw 9 is not an effective and robust piece of legislation.

1.6.3 Speed limits

River Byelaw 48 imposes an 8 knot speed limit on the River Thames above Wandsworth Bridge, the creeks that are linked to the tidal section of the River Thames, and the area off Southend. The section of the Thames from Wandsworth Bridge to Cherry Garden Pier (**Figure 1**) has an advisory speed limit of 12 knots, which was established through the PLA's NTM No.12 of 2009 (**Annex B**). The authority's decision to recommend a speed limit of 12 knots in this area was made following careful consideration of several factors, including the navigable water available, reaction times, navigational hazards, and vessel types.

1.6.4 Revision of byelaws

In 2008, the PLA consulted informally with port and river users on new draft byelaws intended to replace all of the existing River Byelaws with the exception of River Byelaw 9. The PLA intends to revoke River Byelaw 9 when national legislation covering alcohol limits for pleasure users is introduced.

The revised byelaws, known as the Thames Byelaws, include Byelaw 16 – Speed Limits, which introduces a mandatory 12 knot speed limit (with provision of exceptions for certain vessels) between Wandsworth Bridge and Margaretness (**Annex C**). The Thames Byelaws were submitted to the Department for Transport (DfT) for approval in June 2009. The PLA received DfT's response in June 2011 and, following the completion of changes requested by DfT, the draft was resubmitted to the PLA board in July 2011.

In August 2011, the PLA commenced a period of formal public consultation on the draft byelaws. In response, the PLA has received 18 objections, which it is currently addressing. A timeline of the introduction of the Thames Byelaws is shown in **Table 1**.

Table 1 – Timeline of the Introduction of the Thames Byelaws		
2008	June	Informal consultation with users
2009	April	Revised proposals to the PLA Board - agreed
2009	June	Draft submitted to DfT Legal Department (as 'Thames Byelaws') for their consideration (and wider consultation within Government and major stakeholders)
2011	June	DfT legal changes completed
2011	July	Resubmission of amended draft to board
2011	August/September	Formal public consultation for amended byelaws
2011	October	Following public consultation, PLA is currently addressing 18 objections to the drafted byelaws.

1.6.5 Enforcement

The PLA has a fleet of harbour service launches available to enforce the River Byelaws, which are equipped with breathalysers compliant with the Road Traffic Act 1988. In addition, its marine officers are trained in the use and operation of the breathalysers and are permitted to take samples under River Byelaw 9.

The PLA does not enforce River Byelaw 36, which requires vessels to sound one prolonged blast when leaving a berth and entering a fairway primarily because the frequent sounding of whistles in busy areas could lead to uncertainty and confusion between vessels. There are also many residential properties in the vicinity of riverside piers.

1.6.6 Promulgation of information

The PLA uses several methods to publicise the requirements of its byelaws and regulations, and general safety information. In addition to established systems and initiatives, such as river user consultative forums, NTMs, notices to agents, berths and ships' operations, and the PLA website (www.PLA.co.uk), the authority has also introduced other measures. These include:

- A dedicated website (www.boatingonthethames.co.uk) aimed at pleasure boat users that provides navigational advice, including the precautions that should be taken while using the river, and applicable NTMs.
- The publication and distribution of a recreational user guide and codes of practices for both commercial and recreational activities.
- Public meetings with user groups, and close liaison with the many clubs and associations based on the River Thames.
- Presentations at events, such as boat shows, to promote good practice and to raise awareness of the PLA's regulations.
- Broadcasts every 30 minutes by Thames VTS, on VHF channel 14, providing details of works and restrictions in place on the river.

SECTION 2 - ANALYSIS

2.1 AIM

The purpose of the analysis is to determine the contributory causes and circumstances of the accident as a basis for making recommendations to prevent similar accidents occurring in the future.

2.2 THE COLLISION

It is evident from **Figures 5 and 6a, 6b and 6c** that *Sun Clipper* and *Morfil* were potentially in sight of each other for approximately 10 seconds prior to the collision. However, the lack of reaction by *Morfil*'s coxswain indicates that he did not see the oncoming ferry until he took action to try and avoid a collision between 1 and 2 seconds before the impact. By then, the two vessels were too close and closing too quickly for his instinctive attempt to manoeuvre clear to be effective. Although *Sun Clipper*'s master immediately reduced speed when he saw *Morfil* approaching, he was prevented from taking any further avoiding action by the close proximity of the temporary mooring buoys and the No.3 arch buttresses (**Figure 3**).

After the collision, *Morfil*'s owners were extremely fortunate that the RIB, the engine of which continued to run because the coxswain had not fitted a kill cord, did not strike them and cause serious injury – or worse – as it passed close by. They were also fortunate that their rescuers were close at hand, otherwise their chances of survival in the strong flowing river, without lifejackets and while under the influence of alcohol, would have been reduced significantly.

2.3 INFLUENCE OF ALCOHOL

The breath specimen provided by *Morfil*'s coxswain nearly 3 hours after the RIB's collision with *Sun Clipper* indicated that the concentration of alcohol in his breath exceeded 35 microgrammes in 100 millilitres. Assuming that:

- 35 microgrammes equals approximately 3 alcohol units²⁰;
- the concentration of alcohol was only marginally in excess of 35 microgrammes; and
- an average healthy male can metabolize one unit of alcohol per hour

it is likely that *Morfil*'s coxswain had at least 6 units of alcohol in his body system when the vessels collided. This was at least double the alcohol limit prescribed for motorists and professional mariners. Consequently, given that the coxswain's breath specimen was tested 6 hours after he first consumed alcohol at the theatre, he is likely to have consumed in the region of 9 alcohol units during the course of the evening.

The effects of alcohol vary between individuals and depend on a range of factors including: weight, gender, age, metabolism, stress levels and the amount of alcohol consumed. Nevertheless, the effects can generally be considered to impair motor co-ordination and judgment, affect cognitive ability, slow down reaction times, and reduce peripheral and night vision. Alcohol can also affect a person's mood by

²⁰ In the UK, one unit of alcohol is equal to 10 millilitres of pure alcohol. Alcohol concentrations in the human body depend on a number of factors, including gender, body mass, and speed of absorption into the bloodstream.

reducing levels of anxiety, relaxing inhibitions, and increasing their confidence levels. This frequently results in a person being more likely to take risks. Motorists have been found to be 10 times more likely to have an accident when they are 1.5 times over the legal alcohol limit.

In this collision, it is likely that alcohol consumption influenced several of *Morfil's* coxswain's decisions, actions and omissions, including:

- Deciding to take the RIB out for a trip on the River Thames at short notice without any preparation or planning.
- Accepting the risk of operating on the river, at night, without wearing lifejackets.
- Not wearing a kill cord.
- Extending the trip to St Katherine Docks to consume more alcohol.
- Proceeding at a fast speed in the dark without a chart plotter or speedometer to assist with navigation.
- Not noting the vessel's position on the river in relation to key landmarks such as the bridges.
- Not maintaining a VHF radio watch.
- Being slow to react to *Sun Clipper's* presence near the bridge until between 1 and 2 seconds before the collision
- Indecision when taking avoiding action.

Morfil's speed when the RIB collided with *Sun Clipper* cannot be accurately determined. However, given that the boat's engine throttle lever was set to three quarters of full power, and taking into account the estimates of *Sun Clipper's* master and mate, and CCTV footage (**Figures 5, 6a, 6b and 6c**) it is clear that the RIB was travelling significantly in excess of the 12 knot speed limit advised by the PLA. Consequently, *Morfil's* coxswain had only 10 seconds to see *Sun Clipper*, assess her movement, and to take avoiding action.

Nevertheless, given the RIB's manoeuvrability, 10 seconds should have been sufficient time to avoid the collision if *Morfil's* coxswain had not been impaired by alcohol and had he kept a proper lookout ahead. Although there would inevitably have been some luminescence from the lights on the riverbanks, it is unlikely to have been significantly detrimental to the coxswain's night vision. Furthermore, **Figure 10** shows that the Thames Clippers' vessels are well lit and quite conspicuous at night, even under a bridge arch.

2.4 DETERRENCE

Morfil's coxswain was aware that when he set off on his river trip, he had consumed too much alcohol to be able to drive a car without breaking the law. Yet he considered that the consumption of alcohol, and then driving a boat, was acceptable. Such behaviour is not uncommon, with a number of pleasure craft owners viewing boating and drinking as compatible social activities.

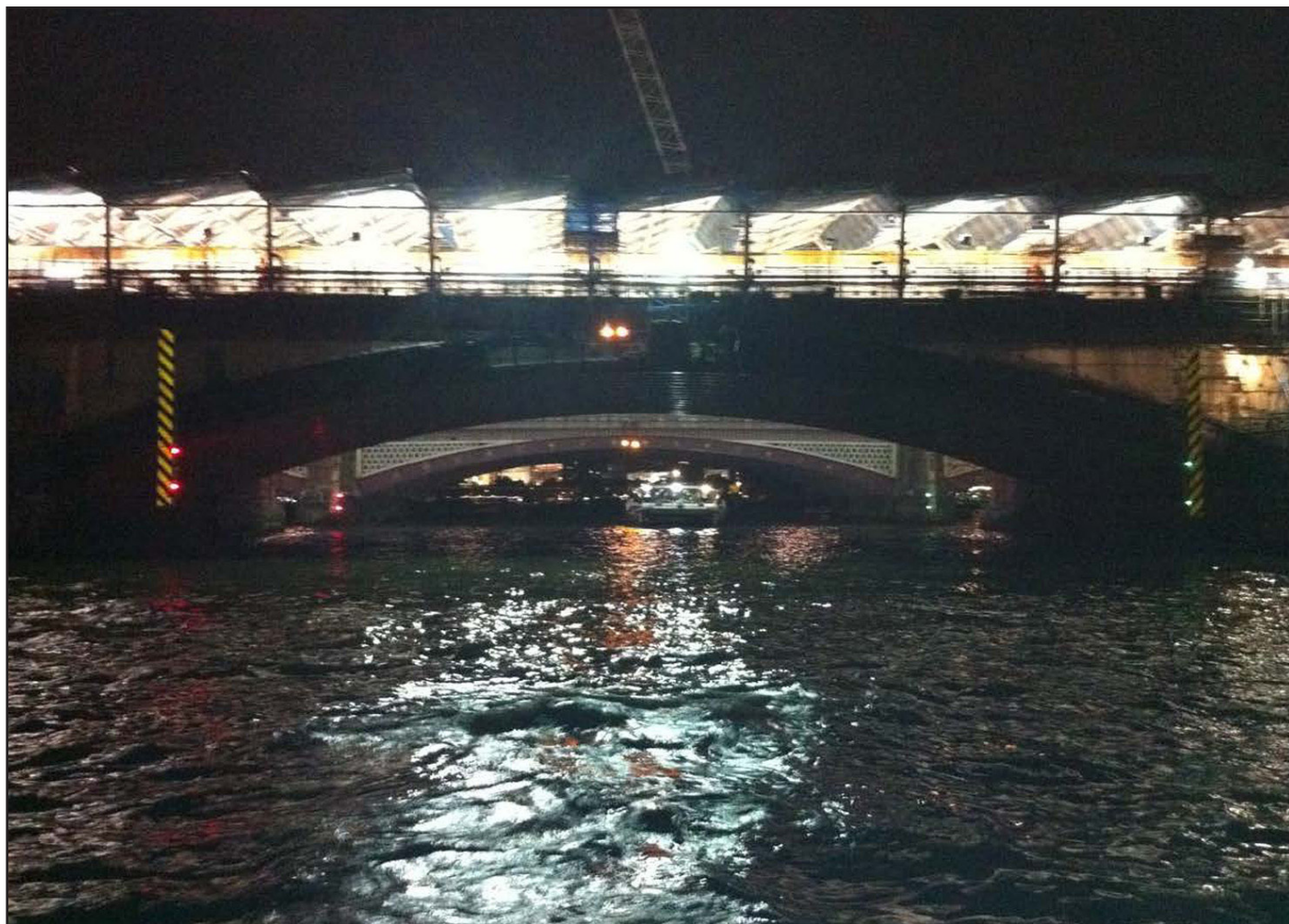


Figure 10: Thames Clipper at Blackfriars Bridges at night

However, it is clear from this and other alcohol-related accidents and fatalities in recent years (**Table 4**), that operating pleasure craft while under the influence of alcohol is dangerous. The need to deter the drivers and coxswains of pleasure craft from consuming alcohol is compelling. While education is an important tool in this respect, the imposition of alcohol limits has a major role to play in changing attitudes towards operating pleasure craft while under the influence of alcohol.

At the moment, local harbour authorities may impose and enforce alcohol limits on pleasure vessel users through byelaws, such as River Byelaw 9. However, notwithstanding the efforts of the RYA to publicise²¹ the existence of byelaws affecting pleasure vessels, it is highly likely that the majority of pleasure vessel users are unaware that such byelaws exist. Moreover, if, as the PLA has experienced: local police forces are reluctant to assist with the enforcement of byelaws; legal advice suggests that prosecutions under these byelaws are not likely to be successful; and, the penalties applicable through byelaws are significantly less than under national legislation (**Table 2**), then byelaws are of little value as a deterrent. The piecemeal application of byelaws by harbour authorities can never be an efficient way of restricting the consumption of alcohol by pleasure vessel users, and the byelaws have no deterrent value whatsoever in waters which lie outside a harbour authority's jurisdiction. The only effective way of deterring excessive alcohol consumption would appear to be through national legislation.

²¹ The RYA website section on regulation draws the attention of the user to Byelaws and Local regulations.

The shortcomings in *Morfil*'s coxswain's observance of official guidance strongly indicate that the PLA must continue to explore innovative methods of promulgating local regulations and navigational advice to its users, possibly including an increased use of signage on the River Thames's banks and bridges. The effective promulgation of information will be pivotal to the safety of vessels during the increased activity on the River Thames expected in 2012.

2.8 SPEED ON THE RIVER THAMES

Rule 6 of the COLREGS (**Annex E**) requires vessels to "*proceed at a safe speed*" in order to enable proper and effective action to be taken to avoid a collision. However, many harbour authorities also impose an upper speed limit for vessels operating in their areas of responsibility. The PLA's use of speed limits to control shipping movements in some of its waters is not unusual.

Although some vessel owners and operators might consider the recommended 12 knot speed limit between Wandsworth Bridge and Cherry Garden Pier (**Figure 1** and **Annex B**) to be unnecessarily restrictive, it was identified as a control measure through risk assessment, and was based on the careful consideration of a number of factors, including reaction times and the proximity of dangers. Indeed, by allowing vessels to close head on at speeds up to 24 knots (12.5m per second) the limit is considerably more generous than the speed limits imposed by many other harbour authorities. In this case, had *Morfil* been adhering to the speed limit, the RIB's coxswain and *Sun Clipper*'s master would have had more time, albeit only seconds, for them to see each other and to take more effective avoiding action.

At present, the 12 knot speed limit is only an advisory limit due to the significant delays the PLA has experienced in introducing the Thames Byelaws (**Table 1**). Similar delays have been experienced by many harbour authorities, largely due to the length of time taken by the DfT to process the port authorities' application for changes.

It is almost certain that the 12 knot speed limit between Wandsworth Bridge and Cherry Garden Pier will further contribute to the safe navigation of vessels on the River Thames when it is made mandatory and enforced. Therefore, should the PLA encounter any further delays in revising its byelaws, particularly the introduction of Thames Byelaw 16 (**Annex C**), it would be prudent for the authority to use other means to make the speed limit mandatory, possibly by issuing a General Direction.

SECTION 3 - CONCLUSIONS

3.1 SAFETY ISSUES DIRECTLY CONTRIBUTING TO THE ACCIDENT WHICH HAVE RESULTED IN RECOMMENDATIONS

1. At the time of the accident, *Morfil's* coxswain is likely to have had at least 6 units of alcohol in his body, over double the alcohol limit prescribed for motorists and professional mariners. It is therefore likely that his decision making and actions were impaired by alcohol. [2.3]
2. The need to deter persons in charge of pleasure vessels from consuming alcohol over defined limits is compelling. [2.4]
3. The current byelaw imposing an alcohol limit for persons in charge of pleasure vessels is not an effective deterrent. [2.4]
4. It is of utmost concern that, despite first being recommended almost 20 years ago, a national alcohol limit for persons in charge of pleasure vessels has not yet been introduced. [2.5]
5. The continued absence of an alcohol limit for persons in charge of pleasure vessels compromises the safety of all water users. [2.5]
6. *Morfil's* coxswain was oblivious to, or ignored, several of the PLA's local regulations and recommendations applicable to vessels navigating on the River Thames. [2.7]
7. The effective promulgation of information will be pivotal to the safety of vessels during the increased activity on the River Thames expected in 2012. [2.7]
8. *Morfil's* speed significantly exceeded the recommended 12 knot limit. Had *Morfil's* coxswain complied with the 12 knot speed limit recommended by the PLA, both he and the master of *Sun Clipper* would have had more time to take more effective avoiding action. [2.8]
9. The 12 knot speed limited recommended by the PLA is unlikely to be effective until it is made mandatory and is enforced. [2.8]

3.2 OTHER SAFETY ISSUES IDENTIFIED DURING THE INVESTIGATION ALSO LEADING TO RECOMMENDATIONS

None

3.3 SAFETY ISSUES IDENTIFIED DURING THE INVESTIGATION WHICH HAVE BEEN ADDRESSED OR HAVE NOT RESULTED IN RECOMMENDATIONS

1. As *Morfil's* coxswain was not using the kill cord effectively, the RIB's crew were extremely fortunate to avoid serious or even fatal injuries from the RIB as it passed close by them while they were in the water. They were also fortunate not to drown in the fast flowing water, despite not wearing lifejackets, because their rescuers were close at hand. [2.2]

2. The vessels did not see each other until about 10 seconds before the collision due to the vessels operating under the No.2 arches and the bridge buttresses. [2.3, 2.6]
3. Although *Sun Clipper* did not sound one prolonged blast on leaving Blackfriars Pier, had the sound signal been made it is unlikely to have been either heard or understood by *Morfil's* coxswain. [2.7]